

STM32 CAN Telemetry Node

Rev A - Bring-Up Test Plan

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Design Tool	Altium Designer
PCB	STM32 CAN Telemetry Node Rev A
MCU	STM32F103C8T6, LQFP-48
Status	Schematic complete - PCB routed - DRC clean - Bring-up planned

This document defines the planned bring-up sequence for the STM32 CAN Telemetry Node Rev A after PCB assembly. The goal is to validate power, programming, sensor inputs, and CAN communication in a controlled order. This is a planned test procedure; physical board validation has not been completed yet.

1. Required Equipment

- Current-limited bench power supply
- Digital multimeter
- Oscilloscope if available
- ST-Link programmer/debugger
- STM32CubeProgrammer and STM32CubeIDE
- CAN analyzer or USB-to-CAN adapter
- Jumper wires and test leads
- Known voltage source or potentiometer for ADC testing
- 120 ohm CAN termination if needed

2. Pre-Power Visual Inspection

Check	Pass/Fail	Notes
Inspect PCB for solder bridges or visible damage		
Verify diode and LED orientation		
Verify STM32 pin 1 orientation		
Verify CAN transceiver orientation		
Verify regulator orientation and solder joints		
Confirm connector labels match schematic intent		
Inspect mounting holes and board edges		

3. Continuity Checks - Board Unpowered

Before applying power, use a multimeter to confirm that the major rails are not shorted to ground.

Measurement	Expected Result	Pass/Fail
VIN_12V to GND	No direct short	
+5V to GND	No direct short	
+3V3 to GND	No direct short	
TP4 and TP7 to GND	Continuity to ground	

4. First Power-Up

- 1 Set bench supply to 12 V.
- 2 Set current limit to a conservative value such as 100-200 mA for first power-up.
- 3 Connect supply to J1 with correct polarity.
- 4 Turn on supply and watch current draw.
- 5 Turn off immediately if current draw is unexpectedly high, any component heats quickly, or visible damage appears.

Item	Expected Result	Pass/Fail
Input current	Stable and below current limit	
Power LED D3	On when +3.3 V is active	
Regulators	No overheating	
Board behavior	No smoke, smell, or visible damage	

5. Power Rail Measurements

Test Point	Rail	Expected Measurement	Pass/Fail
TP1	VIN_12V	Approximately input voltage	
TP2	+5V	Approximately 5.0 V	
TP3	+3V3	Approximately 3.3 V	
TP4	GND	Ground reference	

6. STM32 Programming Test

- 1 Connect ST-Link to J4: +3.3 V, SWDIO, SWCLK, NRST, and GND.
- 2 Open STM32CubeProgrammer.
- 3 Attempt to detect the STM32 device.
- 4 If detected, flash a basic LED blink program.

5 Confirm LED_STATUS toggles.

Test	Expected Result	Pass/Fail
ST-Link connection	STM32 detected	
Flash programming	Program downloads successfully	
LED_STATUS	Blinks under firmware control	

7. Reset and BOOT0 Test

Test	Procedure	Expected Result	Pass/Fail
Reset	Press SW1	MCU resets and firmware restarts	
BOOT0 normal mode	JP1 open/normal setting	MCU boots from flash	
BOOT0 bootloader mode	Install JP1 if needed	MCU can enter system bootloader	

8. CAN Interface Test

- 1 Connect a CAN analyzer or USB-to-CAN adapter to J2.
- 2 Use 120 ohm termination only if the board is at the end of the CAN bus.
- 3 Flash firmware that transmits CAN ID 0x100 at 500 kbps every 100 ms.
- 4 Observe CAN frames on the analyzer.
- 5 Check CANH and CANL with an oscilloscope if available.

Test	Expected Result	Pass/Fail
CAN frame reception	Analyzer receives standard ID 0x100	
DLC	8 bytes	
Payload	Contains AIN1-AIN4 raw ADC values	
LED_CAN	Toggles on transmit	
Bus health	No repeated bus-off or error states	

9. Sensor Input Test

Input	Procedure	Expected Result	Pass/Fail
AIN1-AIN4	Apply known voltages through safe test source	ADC counts change consistently with voltage divider scaling	
DIN1-DIN2	Toggle input between low and valid logic high	GPIO state changes correctly	
FREQ_IN	Apply square wave test signal	Timer input detects frequency	

10. Final System Acceptance Checklist

Acceptance Item	Pass/Fail	Notes
Board powers from 12 V input without abnormal current draw		
+5 V and +3.3 V rails measure correctly		
STM32 programs over SWD		
Status LED operates under firmware control		
AIN1-AIN4 read expected ADC values		
DIN1-DIN2 logic states read correctly		
FREQ_IN timer capture works		
CAN ID 0x100 transmits at 500 kbps		
CAN analyzer receives expected payload		